Fair turns without clashes

Optimization · Level 1 · Lesson 4



In this lesson you will

- Draw clashes as dots and lines.
- Colour the dots so nothing clashes, with as few colours as you can.
- Share turns fairly.

Key ideas

- A line means: not at the same time
- Each colour is a time
- Three tests all joined need three times
- Take turns: nobody always gets the worst

Activity: Colour the clashes

1. Draw (5 min): Draw the eight tests as dots.
2. Join (4 min): Join the tests a pupil takes together.
3. Colour (7 min): Use as few colours as you can.
4. Share (4 min): Compare with a partner.

Use coloured pencils, or write the slot numbers 1, 2, 3 in the dots. No device needed.

Let’s discuss

1. Why do Omar’s three tests need three times?
2. Why does starting with the hardest dot help?
3. Which turn at school would you share more fairly?

Exit ticket

- A line between two dots means ___. The fewest times for our test week is ___.